BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021 • Paper Code: PR XI (01) 21

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 20

Note: Instructions (printed in Urdu on the sheet): 1. Each question has four circles; fill in only the circle of the correct answer. 2. Use a blue or black pen to fill the circles. 3. Filling more than one circle will be treated as a wrong answer.

Q1. $\tan(\alpha - \beta) =$ .....

(A) $(\tan\alpha - \tan\beta)/(1 - \tan\alpha\tan\beta)$

(B) $(\tan\alpha + \tan\beta)/(1 + \tan\alpha\tan\beta)$

(C) $(\tan\alpha + \tan\beta)/(1 - \tan\alpha\tan\beta)$

(D) $(\tan\alpha - \tan\beta)/(1 + \tan\alpha\tan\beta)$

Q2. $\cos(\pi/2 - \theta) =$ .....

(A) $-\sin\theta$

(B) $-\cos\theta$

(C) $\cos\theta$

(D) $\sin\theta$

Q3. If b = 14cm, c = 12 cm and $\alpha = 82°$ the area of triangle is equal to.....

(A) $(bc/2)\cos\alpha$

(B) $(bc/2)\sin\alpha$

(C) $(bc/2)\sin\beta$

(D) $(bc/2)\sin\gamma$

Q4. $\cos\alpha =$ .....

(A) $(b^2 + c^2 - a^2)/2bc$

(B) $(a^2 + b^2 - c^2)/2bc$

(C) $(a^2 + c^2 - b^2)/2bc$

(D) $(b^2 + c^2 - a^2)/2ac$

Q5. The minimum value of $y = 1 + 2\sin\theta$ is:

(A) -1

(B) -3

(C) 1

(D) 3

Q6. $\sin^{-1}x + \cos^{-1}x =$ .....

(A) 2π

(B) π

(C) $\pi/2$

(D) $\pi/3$

Q7. The next term in the 2 + 6 + 18 + ... is:

(A) 24 (B) 36

(C) 54 (D) 72

Q8. If tn = 6n + 5, then T_{n+1} = ...

(A) 6n (B) 6n + 11

(C) 6n + 6 (D) 6n − 5

Q9. (n − 2)!/n! is.....

(A) 1/(n−1) (B) 1/n

(C) 1/(n+1) (D) 1/(n(n−1))

Q10. If A and B are mutually exclusive events, then .....

(A) $P(A \cap B) = P(A) + P(B)$ (B) $P(A \cup B) = P(A) + P(B)$

(C) $P(A \cup B) = P(A) - P(B)$ (D) $P(A \cap B) = P(A) - P(B)$

Q11. The number of terms in the expansion of $(a + b)^n$ is.....

(A) n (B) (n − 1)

(C) (n + 1) (D) n(n − 1)

Q12. The middle term in the expression of $(2x + 5y)^4$ is.....

(A) $600\ x^3y^2$ (B) $120\ xy^2$

(C) $500\ xy^3$ (D) $6\ x^2y^2$

Q13. If $f(x) = x^3 - 1/x^3$ then f(x + 1/x) = ...

(A) −1 (B) 0

(C) 1 (D) x^3

Q14. If $z_1 = 2 + 4i$ and $z_2 = 1 - 3i$, then $z_1 - 3z_2$ = ...

(A) −1 + 5i (B) −1 + 13i

(C) −1 − 5i (D) −1 − 13i

Q15. If Z is any non-zero complex number, then $z + \bar{z}$ = ...

(A) 2Im(z) (B) 2Re(Z)

(C) $2\bar{z}$ (D) $-2\bar{z}$

Q16. If A = [[1,3,1],[-1,2,0],[2,0,-2]], then A_{33} = ...

(A) −1 (B) −5

(C) 1 (D) 5

Q17. Evaluate [[0,1,3],[-1,2,1],[2,1,1]] = ...

(A) −9 (B) −15

(C) 9 (D) 15

Q18. The projection of the vector $2\bar{i} + 3\hat{j} - 2\bar{k}$ on the vector $\bar{i} + 2\hat{j} + 3\bar{k}$ is.....

(A) $-2/\sqrt{14}$ (B) $1/\sqrt{14}$

(C) $2/\sqrt{14}$ (D) $3/\sqrt{14}$

Q19. If A(−4, 6) and B(6, 8), then vector AB =

(A) −2i + 2j (B) 2i + 2j

(C) 10i + 2j (D) −10i + 2j

Q20. Which of the given is an A.P?

(A) 2, 4, 8, 16, ... (B) 3, 6, 9, 12, ...

(C) 3, 7, 11, 15, ... (D) 4, 8, 12, 16, ...

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 50

Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

QII. Answer any TEN parts. Each part carries FIVE marks.

1. Multiply $\sqrt{2} + \sqrt{3}i$ and $2\sqrt{2} - \sqrt{3}i$
2. Prove that $|1+x\ y\ z;\ x\ 1+y\ z;\ x\ y\ 1+z| = 1 + x + y + z$
3. If $a = 3i - 5j$ and $b = -2i - 3j$ then find $|a| - |b|$
4. Find the value of 'k' if $2k + 7, 6k - 2, 8k - 4$ are in A.P. Also, find the sequence.
5. Find the sum $1.2 + 2.3 + 3.4 + ... + 99.100$
6. Find n, d, a_{21}, when $a_1 = -40$, $S_{21} = 210$
7. Find n when ${}^{2n}C_3 : {}^{n}C_2 = 36 : 3$
8. Find the middle term in the expansion of $(3x - x^2/2)^9$
9. Find the inverse function of $f(x) = 4 + \sqrt{2}$
10. Prove that $\sin3\theta/\cos\theta + \cos3\theta/\sin\theta = 2\cot2\theta$
11. Solve the triangle when $a = 120$, $b = 240$ $\gamma = 32°$
12. Prove that $\sin^{-1}(77/85) - \sin^{-1}(3/5) = \cos^{-1}(15/17)$
13. If $a = 30$, $\beta = 50°$, $\gamma = 100°$, find the area of triangle.

SECTION-C

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 30

Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

Q. Attempt any THREE questions. Each question carries equal marks.

QIII.

(a) Solve by Cramer's rule, if $x - y + 2z = 10$; $2x + y - 2z = -4$; $3x + y + z = 7$

(b) For what value of c the vectors are coplanar? If $u = i + j + 2k$, $v = 2i + 3j + k$, $w = ci + 2j + 6k$

QIV.

(a) The harmonic mean of two number is 4 4/5 and the geometric mean is 6. What are the numbers?

(b) In how many ways can letters of the word 'Fasting' be arranged?

QV. Solve by mathematical induction $1.2 + 2.3 + 3.4 + \ldots\ldots + n(n + 1) = n(n + 1)(n + 2)/3$

QVI.

(a) Prove that $\sin3\theta + \sin5\theta + \sin7\theta + \sin9\theta = 4\cos\theta \sin6\theta \cos2\theta$

(b) Show that $r_1 = s \tan(\alpha/2)$